CST570 - Design and Analysis of Algorithms

CST 570 (Design and Analysis of Algorithms) Syllabus

Course Information

Course Details

- **Course Title:** Design and Analysis of Algorithms
- **Course Number:** CST 570
- **Meeting Location:** Physical
- **Units:** 4

Course Description

This course provides an in-depth exploration of advanced data structures and algorithmic techniques, designed for graduate students who have a foundational understanding of data structures and algorithms. The course emphasizes both theoretical analysis and practical implementation of efficient solutions to complex problems. Topics include balanced trees, segment trees, amortized analysis, graph algorithms, network flows, dynamic programming, backtracking, complexity theory (P, NP, NP-completeness), and approximation algorithms. Students will engage in advanced topics and current research in the field.

Materials

- Course textbook: "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein (CLRS)

Technology Requirements

Students must have access to: - A computer with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Analyze algorithm complexity using Big-O, Big-Theta, and Big-Omega notations
2. Apply algorithm design paradigms such as divide-and-conquer, dynamic programming, and greedy approaches
3. Implement advanced data structures and evaluate their efficiency
4. Understand NP-completeness and explore approximation and randomized algorithms

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Projects

Projects include algorithm implementation tasks, complexity analysis, and empirical evaluations of algorithm performance. Students will also conduct research on open algorithmic problems and present findings.

Exams

Exams will cover theoretical analysis, proofs of correctness, and problem-solving related to algorithm design. Exams will include both written problem-solving and analysis of algorithms.

Participation

Class participation involves engaging in discussions, collaborative problem-solving sessions, and contributing to peer reviews. Attendance and active involvement are required.

Grade Assignment

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion. Students will be notified of any substantive changes via Canvas announcement and/or email.